NATIONAL LAND ACQUISITION & MANAGEMENT SYSTEM

Complete Problem Understanding, Solution Design and SIH Implementation Blueprint

Purpose: A super-explanatory guide to understand the SIH problem statement, convert it into a practical product, and design a demonstrable prototype.

Primary audience: SIH team members, developers, UI/UX designers, AI/ML members and presenters.

Core idea: Digitize and monitor the entire land acquisition lifecycle in one secure, GIS-enabled national platform.

Problem -> Workflow -> Platform -> GIS -> AI -> Dashboard -> Implementation

Contents

1. The Problem in One Minute

The government often needs to acquire land before a public project can be executed. Examples include highways, railways, industrial corridors, irrigation systems, airports, urban development and renewable-energy projects. Land acquisition is not one action; it is a chain of activities involving multiple departments and large amounts of information.

The SIH problem statement is essentially asking for one national digital platform that can track this chain from the moment a project asks for land until the government obtains possession and completes the required rehabilitation and resettlement activities.

```
CURRENT SITUATION
Many departments + paper/Excel/PDF + separate portals + manual coordination
                         |
                         v
              Delays / duplication / poor visibility

PROPOSED SITUATION
One platform + workflow + GIS + documents + dashboards + APIs + analytics
                         |
                         v
              Transparent, traceable, measurable acquisition
```

The simplest interpretation: Instead of asking several offices for the current status of a project, an authorized officer should be able to open the platform, select the project or parcel, and see its current status immediately.

2. Why Land Acquisition Is Hard

The difficulty is not simply that there are many forms. The deeper problem is coordination. A single project can involve a central ministry, a state department, district officials, revenue records, survey information, landowners, compensation, legal/administrative decisions, field verification and rehabilitation measures.

- Different stakeholders may have different responsibilities and access rights.
- Different states or agencies can follow different administrative workflows.
- Land is spatial data, so location and parcel boundaries matter as much as text records.
- Documents are critical evidence and must remain traceable over time.
- Compensation is not the same as acquisition: money assessed, approved, paid and possession taken are different events.
- Rehabilitation and resettlement can continue even after a parcel has been acquired.
- Delays may happen at one stage while the project dashboard still needs to show the exact bottleneck.

3. What the SIH Problem Statement Is Actually Asking You to Build

It is not asking for a normal website with a few forms. The intended solution is closer to a government workflow and monitoring platform.

Capability	Meaning in simple language
Workflow management	Move an acquisition case from one official/stage to another with status, approvals, deadlines and audit history.

Capability	Meaning in simple language
Digital proposal processing	Allow project agencies to submit land requirements online instead of relying on physical files.
GIS	Show land parcels and project corridors on an interactive map.
Document repository	Store evidence and supporting documents in a structured, searchable place.
Real-time monitoring	Know how much land is proposed, notified, acquired and possessed right now.
MIS dashboards	Give project, state and national level summaries to decision-makers.
API integration	Exchange data with existing land-record and government systems rather than recreating everything.
Mobile field collection	Let field officers verify parcel information at the physical site.
Predictive analytics	Use data to identify likely delays, risk and inconsistencies before they become bigger problems.

4. Land Acquisition Lifecycle Explained Step by Step

The exact legal procedure can vary by project and governing framework. For the software design, however, the lifecycle can be represented as a configurable sequence of stages.

Project Proposal: A project implementing agency states the project objective, geography, expected land requirement, tentative timeline and supporting documents.

Land Requirement Identification: Required areas are identified by district, village, survey number and parcel. The system stores proposed area and location.

GIS Mapping: Land parcels are plotted or imported on a map. The platform can store polygon geometry, coordinates and spatial metadata.

Verification and Scrutiny: Authorized officials verify records, survey information, document completeness and potential inconsistencies.

Approval: The proposal is routed to the appropriate authority. Every decision is recorded with date, officer and remarks.

Notification: When applicable, notification-related information and dates are entered into the system and linked to affected parcels/families.

Claims/Objections/Case Handling: Any relevant claims, objections or issues are recorded and assigned for review.

Compensation Assessment: Compensation for eligible landowners/affected parties is calculated, reviewed and recorded.

Award Declaration: The award and supporting records are uploaded and linked to the affected parcel/case.

Compensation Disbursement: The system tracks assessed, approved, initiated and paid amounts, along with payment references where available.

Possession: Possession events are recorded parcel-wise, including date and supporting evidence.

R&R: Affected/displaced families, entitlements and rehabilitation/resettlement progress are tracked.

Closure: The case/project stage is marked complete only when the relevant workflow conditions are met.

```
PROJECT -> LAND IDENTIFICATION -> VERIFICATION -> APPROVAL -> NOTIFICATION
        -> COMPENSATION -> AWARD -> PAYMENT -> POSSESSION -> R&R -> CLOSURE
```

5. Stakeholders and Their Roles

Stakeholder	What they need from the system
Central Ministry / Central Authority	National visibility, policy-level analytics, state comparisons, high-risk project monitoring.
State Government	State-wide progress, approval queues, district performance, escalations and consolidated reporting.
District Administration	Parcel-level work, verification, case management, compensation and possession tracking.
Land Acquisition Officer	Process cases, verify documents, issue/record actions, update statuses and manage deadlines.
Project Implementing Agency	Submit projects, define land requirements, track acquisition and identify project risks.
Revenue / Land Records Team	Verify land record data, survey details and parcel information through approved integrations.
Field Officer	Capture GPS, photos, verification observations and field evidence through mobile interface.
Senior Decision Maker	See dashboards, delays, bottlenecks, trends and risk alerts without reading every file.

6. Scope of the Proposed Platform

The platform should cover both transaction processing and management information. A useful way to define scope is to separate operational features from monitoring features.

Scope area	Included capabilities
Operational	Proposal submission, verification, routing, approvals, notifications, compensation, award, possession, R&R.
Spatial	Map, parcels, project corridors, boundaries, geo-tagging, spatial search and map-based status.
Documents	Upload, versioning, metadata, access permissions, audit history and linkage to cases/parcels.
Monitoring	KPIs, timelines, milestones, pending actions, delays, state/district/project comparisons.
Integration	Land records, cadastral maps, relevant portals and services through documented APIs where available.
Field	Mobile verification, photo evidence, GPS capture and offline-first synchronization as a future-ready feature.
Analytics	Descriptive dashboards plus predictive risk/delay indicators.

7. Major Functional Modules

7.1 Authentication and Role-Based Access Control - Every user logs in according to role. Permissions determine what the person can view, create, edit, approve or export.

7.2 Project Management - Create projects, record project type, implementing agency, geography, schedule, land requirement and responsible authorities.

7.3 Land Parcel Management - Create or import parcel records, area, survey identifiers, location, geometry, status and links to documents/cases.

7.4 Workflow Engine - Route a case to the correct stage/authority, enforce required fields, capture decisions and deadlines, and support escalation.

7.5 Notification Management - Record notifications, dates, affected parcels/families and relevant document references.

7.6 Compensation Management - Track amount assessed, approved, paid and pending, with parcel/family level traceability.

7.7 Possession Management - Record possession date, parcel, evidence and current possession state.

7.8 R&R Management - Track affected/displaced families, entitlements, resettlement activities and pending cases.

7.9 Document Management - Secure repository, metadata, version history, linkage and audit trail.

7.10 Reporting and Dashboards - National, state, district and project-level visualizations and MIS exports.

7.11 Alerts and Escalation - Deadline warnings, overdue flags and hierarchical escalation.

7.12 Integration Layer - API connectors, data validation, synchronization logs and error handling.

8. GIS and Land Parcel Management

GIS is not just a map placed on the homepage. It is a core data model for the system because land is inherently spatial. A parcel should be linked to geometry as well as business information.

Parcel example: Parcel P-101 | Survey No. 123/2 | Area 2.40 acres | District X | Status: Acquired | Compensation: Paid | Possession: Taken

Useful GIS functions include:

- Pan/zoom and base maps
- Parcel search by project, district, village or survey number
- Map layers for proposed/notified/acquired/possessed land
- Click parcel to open its full record
- Draw or import project corridors and parcel boundaries
- GPS capture from mobile field app
- Spatial queries such as parcels within a project corridor
- Map-based filtering by acquisition status

```
MAP
  |-- Project Corridor
  |-- Parcel A [PROPOSED]
  |-- Parcel B [NOTIFIED]
```

```
|-- Parcel C [ACQUIRED]
|-- Parcel D [POSSESSION TAKEN]
```

Recommended spatial database: PostgreSQL with PostGIS, because the solution needs structured business data plus spatial geometry and spatial queries.

9. Workflow Automation and Approvals

A workflow engine prevents the platform from becoming a collection of disconnected forms. Each case has a current stage, assigned responsibility, required actions, deadlines and history.

```
Case ID LA-2026-001
Current Stage: COMPENSATION ASSESSMENT
Assigned To: District Land Acquisition Officer
Due Date: 30-Aug-2026
Status: PENDING

[Complete] [Send Back] [Reject] [Approve / Forward]
```

- Mandatory fields before moving to the next stage
- Automatic assignment based on state/district/project rules
- Approval and rejection with remarks
- Return-to-correction workflow
- Deadline calculation
- Escalation when overdue
- Complete action history

Why this matters: The system should know not only that a case exists, but also where it is stuck, who owns the next action and how long it has been waiting.

10. Compensation, Awards and Possession

These concepts must be modeled separately. A project can have land acquired but compensation pending, or compensation paid for a parcel while possession is not yet recorded. Combining them into one “acquired = yes/no” field would lose important operational information.

Parameter	Example meaning
Compensation assessed	The amount calculated as due.
Compensation approved	The amount accepted by the competent authority.
Compensation paid	The amount actually disbursed/recorded as paid.
Award declared	Award record and date linked to the case/parcel.
Possession status	Not taken / partial / taken, as appropriate to the workflow.
Possession date	Date on which possession event is recorded.
Pending amount	Assessed/approved amount not yet paid, subject to the applicable workflow.

11. Rehabilitation and Resettlement (R&R)

Acquiring land does not automatically mean the human impact is complete. The platform therefore needs a separate R&R area to track affected and displaced families and the progress of rehabilitation/resettlement measures.

- Family/case identification
- Affected vs. displaced status
- Entitlement/benefit category
- Required R&R action
- Action assigned to responsible authority
- Completion date
- Pending reason
- Document/evidence attachment
- R&R progress at project and district level

```
PROJECT X
Affected Families: 127
Displaced Families: 34
R&R Completed: 26
R&R Pending: 8

Dashboard should expose this separately from land acquisition percentage.
```

12. Document Management and Audit Trail

Documents are evidence. The platform therefore needs more than a generic file-upload button.

Requirement	Why it matters
Metadata	Identify document type, date, owner/case/parcel and version.
Version control	Know which version is current and preserve previous versions where policy permits.
Access control	Restrict sensitive documents to authorized roles.
Audit history	Record who uploaded, changed, approved or downloaded the document.
Validation	Check file type, size and optionally integrity/signature requirements.
Search	Find documents by project, parcel, survey number, document type or date.

13. Dashboards, MIS and Monitoring

Dashboards are the decision-making layer. They should not merely display colorful charts; each number should lead to actionable detail.

```
NATIONAL DASHBOARD
Projects: 1,248
Land Proposed: 8.2M acres
Land Acquired: 5.7M acres
Compensation Assessed: Rs. 24,500 Cr
```

```
Compensation Paid: Rs. 21,900 Cr
Affected Families: 2,45,000
Displaced Families: 48,000
High-Risk Projects: 83
Overdue Cases: 412
```

Important drill-down path:

```
NATIONAL -> STATE -> DISTRICT -> PROJECT -> LAND PARCEL -> CASE / FAMILY ->
DOCUMENT / EVENT
```

14. Alerts, Escalation and Timeline Management

The system should proactively tell users about problems rather than waiting for them to search for problems.

Trigger	Possible system action
Deadline approaching	Send reminder to responsible officer.
Deadline missed	Mark case overdue and notify officer.
Repeated delay	Escalate to higher authority.
Required document missing	Block stage completion and notify applicant.
Data mismatch found	Create review flag.
High ML risk score	Surface project in risk dashboard.

15. Government API Integration

The problem statement expects interoperability. That means the platform should be designed as a system that can exchange information with existing government systems, not a closed database that forces every department to re-enter everything.

```
YOUR PLATFORM
     |
     +-- Land Records Connector
     +-- Cadastral Map Connector
     +-- Payment / Financial Connector
     +-- Identity / Verification Connector
     +-- Other approved government services
     |
 Integration Layer: authentication + validation + logging + retries
```

For SIH: If real government APIs are not provided to the team, create realistic mock APIs and clearly label them as mock/integration-ready. Demonstrate request, response, validation and synchronization behavior.

16. Mobile Field Application

Field work is where the digital record meets the physical land. A mobile application can reduce manual transcription and improve evidence collection.

- Secure field login
- Assigned parcel list
- GPS capture
- Parcel/record lookup

- Photo capture
- Verification checklist
- Document/photo upload
- Remarks and issue flags
- Offline storage with later synchronization
- Timestamp and audit information

```
FIELD VISIT
Open assigned parcel -> capture GPS -> verify boundary/record -> capture photos
-> fill checklist -> submit -> sync to central system -> officer dashboard updates
```

17. AI/ML Opportunities That Actually Make Sense

AI should solve an operational problem. Adding a chatbot only because the project mentions AI is weak. The strongest AI features for this use case are those that predict risk, detect inconsistencies or extract structured information from documents.

17.1 Delay Prediction

Use historical project/case data to predict whether a project or workflow stage is likely to exceed its target timeline.

```
Inputs: project size + affected families + disputes + pending compensation + prior
delays + stage duration
                         |
                         v
                ML model (classification/regression)
                         |
                         v
            Delay probability = 71%
            Risk level = HIGH
```

17.2 Acquisition Risk Score

Create a transparent risk score from structured indicators. The dashboard can show why a project is high risk instead of presenting an unexplained black-box number.

- Number of unresolved issues
- Age of pending cases
- Percentage of compensation pending
- Number of affected/displaced families
- R&R completion
- Historical district/project delay
- Data inconsistencies

17.3 Document Intelligence and OCR

An uploaded document can be processed to extract relevant fields. The extracted values can then be compared with existing records.

```
Uploaded Survey Document
          |
          v
```

```
        OCR / NLP
            |
            v
Survey No = 123/2
Area = 2.30 acres
Village = ABC
            |
            v
Compare with Land Record
            |
            v
Area in record = 2.50 acres
            |
            v
FLAG: POSSIBLE MISMATCH
```

17.4 Duplicate and Inconsistency Detection

Rules or ML can flag suspicious duplicates such as the same survey number appearing with conflicting owner, area or project assignments.

17.5 Natural Language Analytics

An authorized officer can ask questions about data stored in the system. The assistant should generate answers from structured database queries and approved analytics, not invent facts.

```
Officer: "Show UP projects delayed by more than 30 days."
System: runs validated query -> returns matching projects -> summarizes the
results.
```

18. Data Model and Key Entities

A strong database design is critical because nearly every module is connected to the same project/parcel/case information.

Entity	Important example fields
User	user_id, role, department, state, district, status
Project	project_id, project_name, agency, state, district, start_date, target_date, status
Parcel	parcel_id, survey_no, area, district, village, geometry, status
Acquisition Case	case_id, project_id, parcel_id, current_stage, assigned_to, due_date, status
Notification	notification_id, case_id, date, type, document_id
Award	award_id, case_id, amount, date, document_id
Compensation	compensation_id, case_id/family_id, assessed, approved, paid, status
Possession	possession_id, parcel_id, date, status, evidence_document
Family	family_id, project_id, affected_status, displaced_status, R&R status
R&R Activity	activity_id, family_id, type, due_date, completion_date, status
Document	document_id, entity_type, entity_id, type, version, storage_path, uploaded_by

Entity	Important example fields
Workflow Event	event_id, case_id, action, actor, timestamp, remarks
Risk Score	risk_id, project_id/case_id, score, factors, model_version, created_at

19. System Architecture

```
                       USERS
             +-----------------------------+
             | Web Portal | Mobile App    |
             +-------------+---------------+
                           |
                  API Gateway / Auth
                           |
      +--------------------+--------------------+
      |                    |                    |
 Workflow Service      GIS Service        Document Service
      |                    |                    |
      +--------------------+--------------------+
                           |
                 Core Backend Services
                           |
      +--------------------+--------------------+
      |                    |                    |
  PostgreSQL            PostGIS            Object Storage
      |                    |                    |
      +--------------------+--------------------+
                           |
               Analytics / AI Services
                           |
              External Government APIs
```

A practical implementation can start as a modular monolith for the SIH prototype. Microservices are not mandatory for a prototype; introducing them too early can increase complexity without improving the demonstration.

20. Suggested Components-wise Technology

Component	Suggested technology	Reason
Frontend	React / Next.js	Fast development of dashboards, forms and role-based views.
UI styling	Tailwind CSS	Consistent responsive UI with rapid prototyping.
Backend	Node.js + Express or NestJS	Good API ecosystem and suitable for workflow-heavy applications.
Primary database	PostgreSQL	Strong relational model for projects, cases, users, finance-related fields and audit data.
Spatial database	PostGIS	Stores and queries parcel geometry and spatial relationships.
Maps	Leaflet or MapLibre	Interactive web maps without locking the application to one UI provider.

Component	Suggested technology	Reason
Authentication	JWT/OAuth-compatible flow + RBAC	Secure authentication and granular permissions.
Object storage	S3-compatible storage	Scalable document and photo storage.
AI service	Python + FastAPI	Separates ML workloads from main application logic.
ML	Scikit-learn / XGBoost	Practical for tabular delay/risk prediction.
OCR	Tesseract or cloud OCR	Extract text/fields from scanned documents.
Mobile	React Native or Flutter	Shared mobile development for field workflows.
Notifications	In-app + email/SMS integration	Alerts and escalation.
Deployment	Docker + cloud / government-approved infrastructure	Repeatable deployment and easier scaling.
API docs	Swagger / OpenAPI	Clear interface documentation and testing.

21. Security, Privacy and Governance

Because the platform could contain land, personal, financial and administrative information, security is a core architecture requirement, not an add-on.

- Least-privilege role-based access
- Encryption in transit and at rest where supported
- Strong password/session controls and, where appropriate, multi-factor authentication
- Immutable or protected audit logs for sensitive actions
- Document-level authorization
- Input validation and secure API design
- Backups and disaster-recovery planning
- Data retention and deletion rules aligned with government policy
- Secure handling of personal information
- Monitoring for suspicious access and repeated failed attempts

22. End-to-End Example

Consider a fictional highway expansion project in Uttar Pradesh requiring 500 acres across several villages.

Stage	What happens in the platform
Proposal	Agency submits project details and requests 500 acres.
Land mapping	Parcels are imported/created and displayed on GIS map.
Verification	Officials verify survey details and flag mismatched documents.
Notification	Notification information is recorded and linked to affected parcels.
Families	Affected/displaced families are recorded for relevant cases.
Compensation	Assessed, approved and paid amounts are tracked.
Award	Award documents are linked to the relevant cases.

Stage	What happens in the platform
Possession	390 acres show possession taken; 110 acres remain pending.
R&R	26 of 34 displaced families have completed relevant R&R actions.
Risk	ML model flags project with high delay probability because of pending cases and historical delays.
Dashboard	Senior officials see project progress and drill down to delayed parcels/cases.

23. SIH MVP: What to Build and What Not to Build

The biggest risk is scope explosion. The official problem statement contains many requirements because it describes the target system. Your SIH prototype does not need a production-scale implementation of every possible integration.

Build this end-to-end

- Role-based login
- Create project
- Add/import a small set of sample parcels
- Interactive GIS map
- Parcel detail page
- Workflow from proposal to possession
- Document upload/version example
- Compensation tracking
- R&R tracking
- Dashboard with filters and drill-down
- Deadline alerts
- One strong AI feature such as delay prediction or document mismatch detection
- Mock external API integration

Do not overbuild for the prototype

- A giant microservice architecture with no working business flow
- A generic chatbot disconnected from project data
- Dozens of unfinished dashboards
- Fake claims of integration with government systems you cannot access
- Complex blockchain merely because it sounds innovative
- A huge AI model with no meaningful dataset or explanation

24. Demo Flow for Judges

Your demonstration should tell a story rather than display isolated screens.

1. Login as a District Land Acquisition Officer.
2. Open the project dashboard and select a highway project.
3. Show the map with proposed/notified/acquired parcels.
4. Click one parcel and show its record, documents, compensation and possession status.

5. Open the workflow and show who is responsible for the next action and the deadline.
6. Show a mismatch detected between a document and land record.
7. Show the compensation dashboard and pending amount.
8. Show R&R progress for affected/displaced families.
9. Switch to senior dashboard and show state/project comparison.
10. Show AI risk score and explain the factors contributing to the risk.
11. Trigger or demonstrate an overdue alert and escalation.

The central message should be: “One platform gives the government a traceable, map-based and measurable view of the entire acquisition lifecycle.”

25. Expected Impact and Success Metrics

Impact area	How success can be measured in the prototype
Transparency	Every major workflow action has a visible status and audit trail.
Time reduction	Compare manual-like workflow steps with automated routing/notifications.
Data consistency	Count and display detected mismatches/duplicate records.
Monitoring	Officials can get project status through dashboard filters rather than manual consolidation.
Field efficiency	Field verification can capture GPS/photos directly into the parcel record.
Decision support	High-risk or overdue projects are surfaced automatically.
Coordination	Multiple roles use the same case record rather than exchanging disconnected copies.
Scalability	Architecture separates spatial, document, workflow and analytics concerns so additional states/projects can be added.

26. Final Solution Summary

The proposed National Land Acquisition & Management System should be treated as a single source of truth for land acquisition. It connects people, land parcels, workflow stages, documents, compensation, possession and rehabilitation/resettlement information in one controlled system.

```
                            NATIONAL LAND ACQUISITION PLATFORM
                                          |
              +---------------------------+---------------------------+
              |                           |                           |
           WORKFLOW                      GIS                      DOCUMENTS
              |                           |                           |
          approvals / deadlines    parcel maps / GPS      versions / audit trail
              |                           |                           |
              +---------------------------+---------------------------+
                                          |
                                 PROJECT DATA PLATFORM
                                          |
                        +-----------------+-----------------+
                        |                                   |
```

```
     DASHBOARDS                              AI / ML
          |                                     |
national/state/project             risk / delay / mismatch
          |                                     |
          +------------------+------------------+
                             |
              BETTER DECISION-MAKING
```

ONE-LINE PITCH

A centralized, GIS-enabled platform that digitizes the complete land acquisition lifecycle, automates inter-department workflows, tracks every parcel and affected family, and uses analytics and AI to identify risks, inconsistencies and delays.

Important implementation note: The software workflow should be configurable because the exact administrative/legal process can differ by project, authority and applicable rules. The prototype should demonstrate a representative workflow rather than claim to implement every statutory requirement.